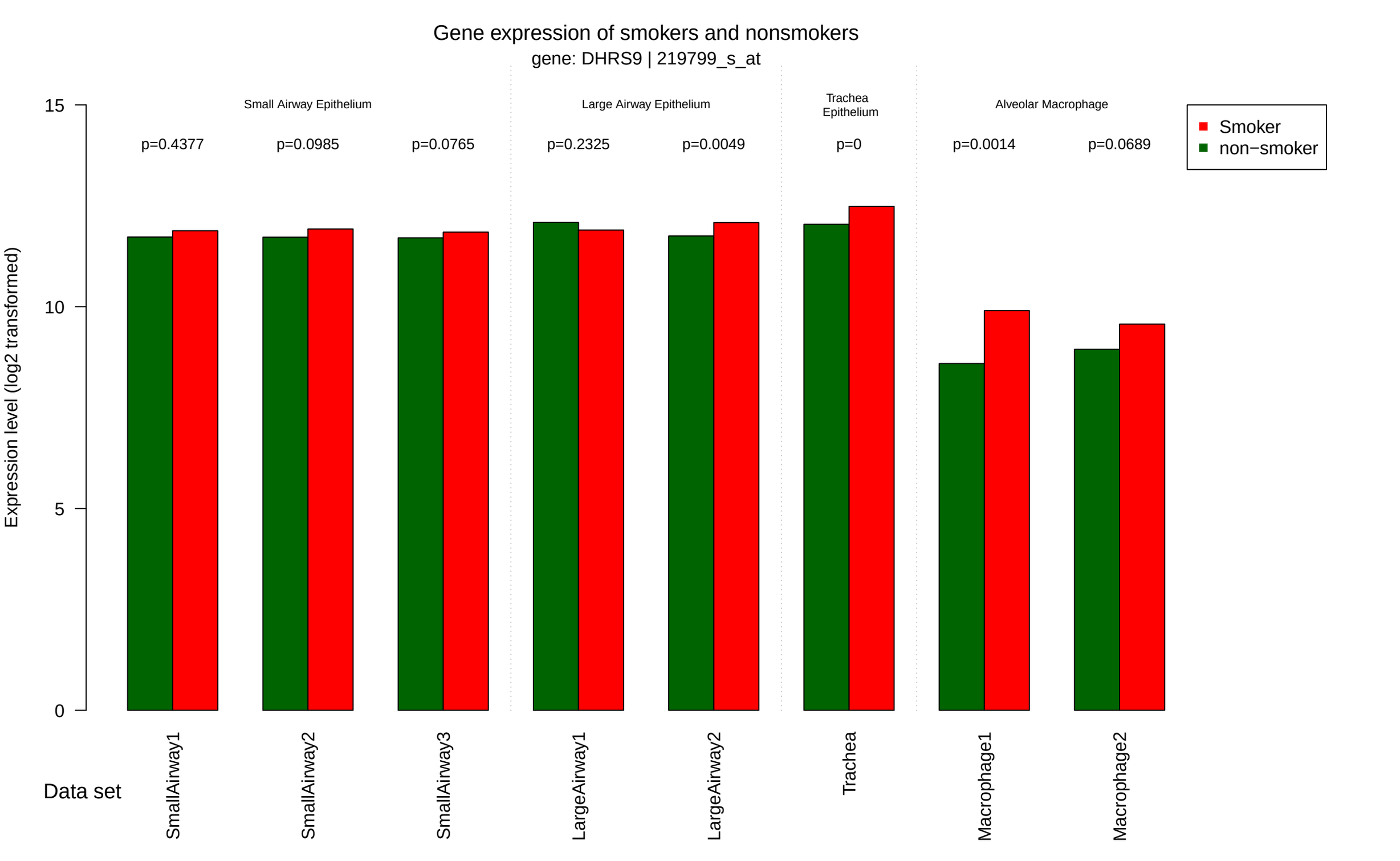
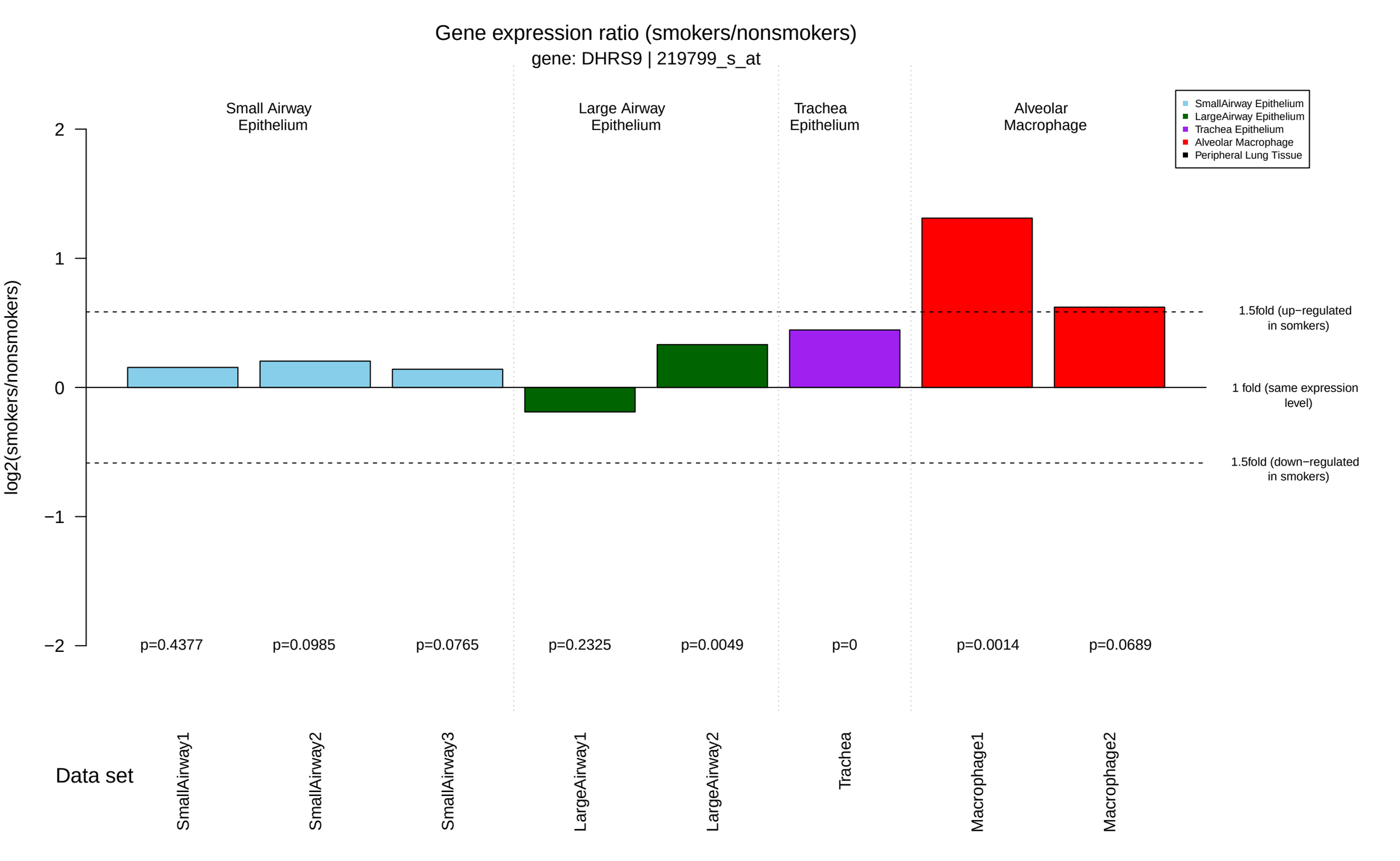
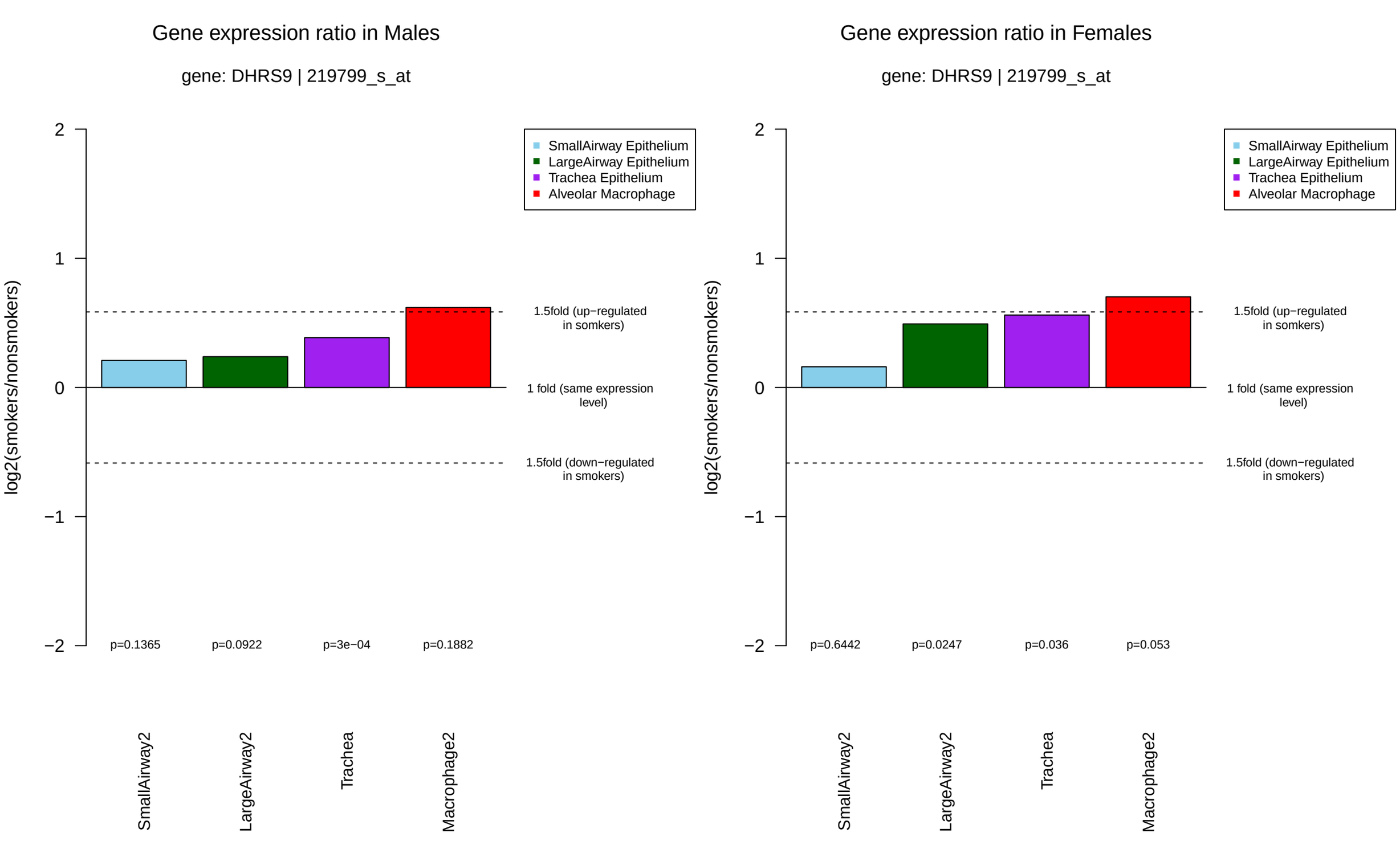
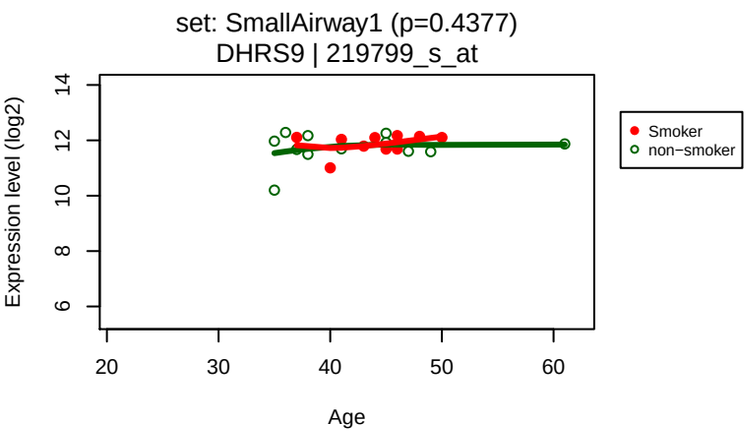
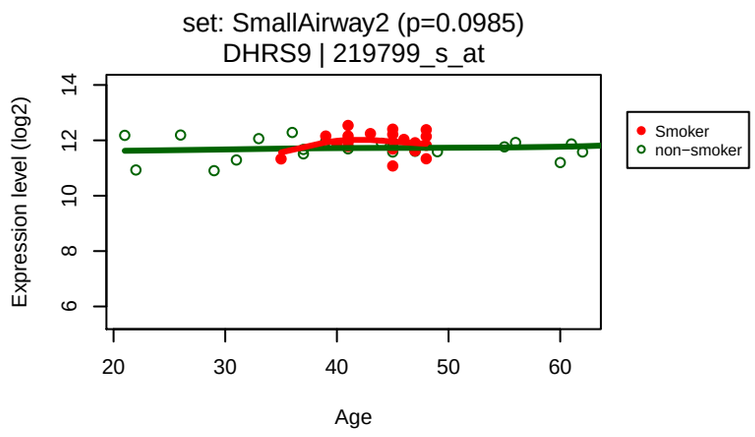
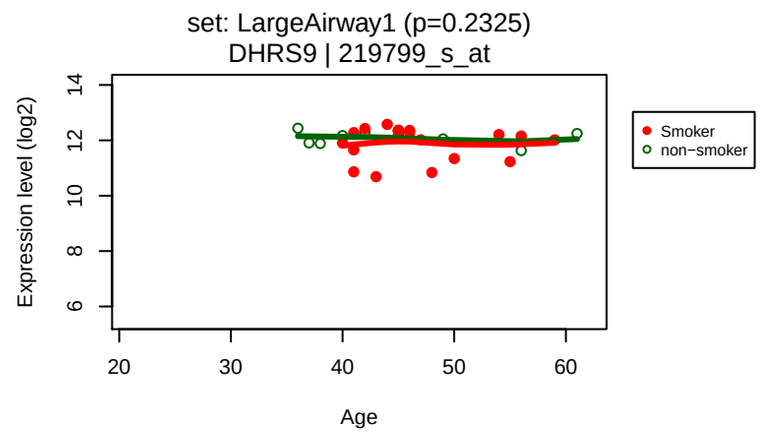

Gene expression of smokers and nonsmokers
gene: DHRS9 | 219799_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4377
p=0.0985
p=0.0765
p=0.2325
p=0.0049
p=0
p=0.0014
p=0.0689
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

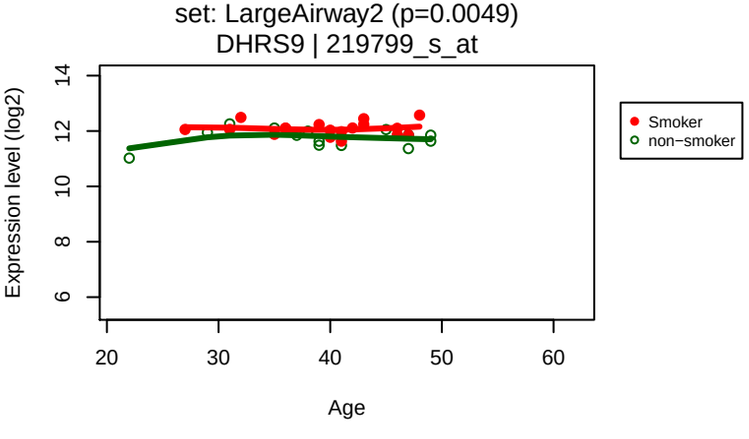

Gene expression ratio (smokers/nonsmokers)
gene: DHRS9 | 219799_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4377
p=0.0985
p=0.0765
p=0.2325
p=0.0049
p=0
p=0.0014
p=0.0689
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

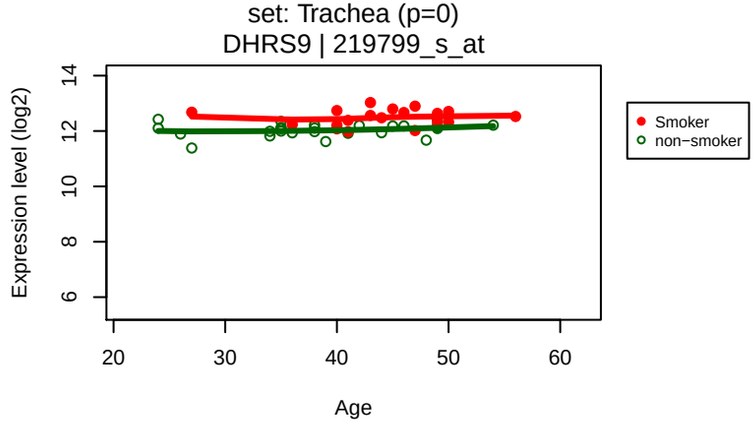

Gene expression ratio in Males
gene: DHRS9 | 219799_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1365
p=0.0922
p=3e−04
p=0.1882
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DHRS9 | 219799_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6442
p=0.0247
p=0.036
p=0.053
SmallAirway2
LargeAirway2
Trachea
Macrophage2

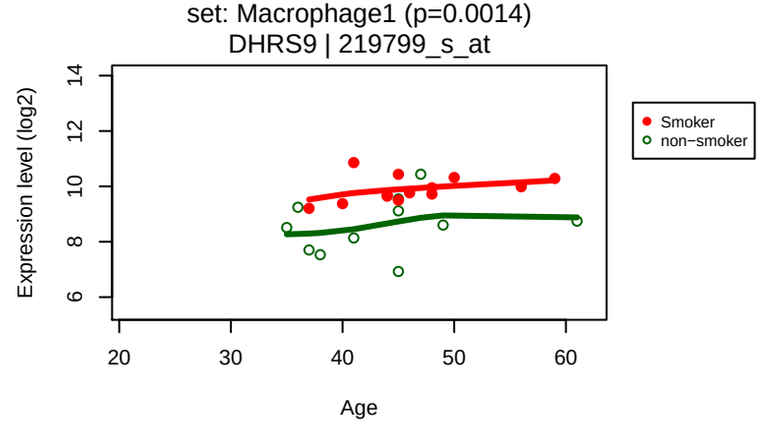

set: SmallAirway1 (p=0.4377)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker

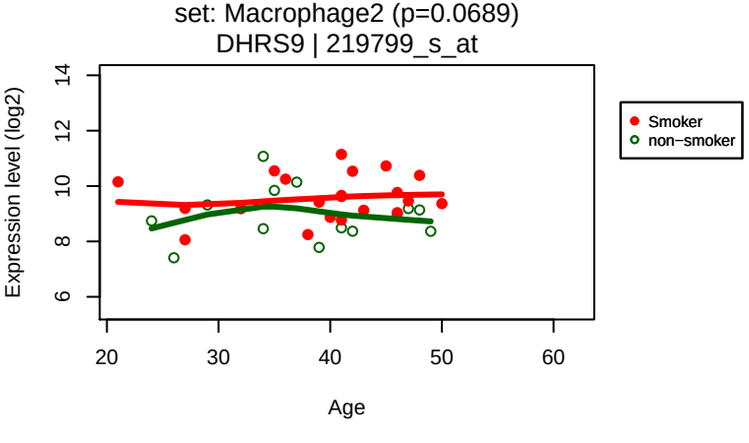

set: SmallAirway2 (p=0.0985)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker

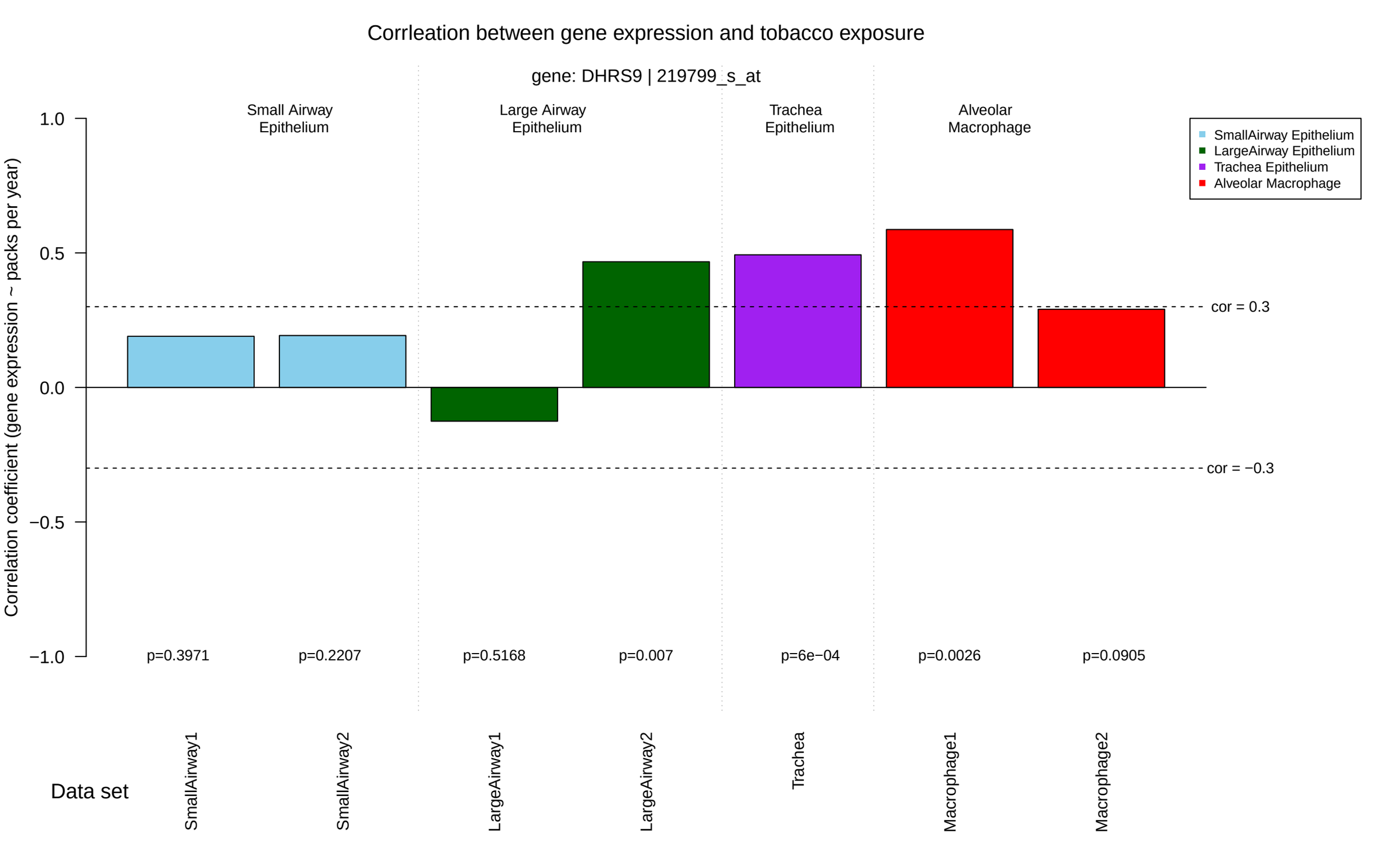

set: LargeAirway1 (p=0.2325)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0049)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0014)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0689)
DHRS9 | 219799_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DHRS9 | 219799_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3971
p=0.2207
p=0.5168
p=0.007
p=6e−04
p=0.0026
p=0.0905
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2